

PROJECT DOCUMENTATION

AI-Based Virtual Try-On System

Summer Internship 2026

Prepared By: Praneetha Ayyana Somayajulu

Role: AI/ML Intern

Date: 05/06/2026

Introduction

This document summarizes the work performed, knowledge gained, and experiences acquired during the Raritone Summer Internship 2026. Throughout the internship, I worked on research and exploration related to Artificial Intelligence, Computer Vision, Virtual Try-On Systems, Human Body Reconstruction, and 3D Avatar Generation.

The internship provided exposure to emerging technologies used in modern fashion technology applications and helped develop both technical and research-oriented skills.

My Role as an AI/ML Intern

As an AI/ML Intern, my primary responsibility was to research and understand technologies used in AI-powered Virtual Try-On systems.

My work included:

- Studying Virtual Try-On technologies.
- Exploring body segmentation techniques.
- Understanding pose estimation methods.
- Researching garment fitting approaches.
- Investigating 3D avatar generation.
- Comparing human body reconstruction models.
- Organizing datasets and digital assets.
- Creating workflow documentation.
- Analyzing future applications of AI in fashion technology.

Research Areas Explored

During the internship, research was conducted in several areas of Artificial Intelligence and Computer Vision.

2D Virtual Try-On Systems

I explored how clothing images can be virtually placed on users using image processing and pose estimation techniques. The research focused on garment alignment, clothing placement, and body landmark detection.

Pose Detection

I studied pose estimation technologies used to identify important body landmarks required for Virtual Try-On applications.

Technologies explored:

- MediaPipe
- OpenPose
- Detectron2

Body Segmentation

I researched body segmentation methods that separate the user from the background and identify body regions for garment fitting.

3D Avatar Generation

I explored techniques used to create realistic digital human representations from images and body information.

Technologies studied:

- SMPL
- PIFuHD
- ICON
- HumanNeRF

- 3D Gaussian Splatting

Comparison Table:

Technology	Input	Output	Advantages	Limitations	Virtual Try-On Use
SMPL	Body Shape & Pose Parameters	3D Human Mesh	Lightweight, widely used	Limited clothing detail	Body Modeling
PIFuHD	Single RGB Image	Detailed 3D Human Model	High-quality reconstruction	High computational cost	Avatar Generation
ICON	RGB Image	Clothed Human Mesh	Better clothing reconstruction	Complex architecture	Virtual Fashion
HumanNeRF	Multi-view Images	Neural Human Representation	Realistic rendering	Requires multiple views	Digital Humans
Gaussian Splatting	Multi-view Images	3D Reconstruction	Fast rendering, high quality	Advanced optimization	Real-time Visualization

Practical Learning Experience

Throughout the internship, I gained practical exposure to various tools and technologies.

Blender Exploration

Blender was explored to understand avatar visualization and asset management. I learned how to:

- Import avatar models.
- Navigate 3D environments.
- Explore avatar structures.
- Visualize 3D assets.

- Understand basic garment placement concepts.

Dataset Management

I learned how to organize and maintain structured datasets for AI research projects.

Dataset categories included:

- Clothing images
- Shoes
- Accessories
- Male avatars
- Female avatars
- 3D assets

Skills Developed

The internship helped me strengthen the following skills:

Technical Skills

- Artificial Intelligence
- Computer Vision
- Machine Learning Research
- Pose Detection
- Body Segmentation
- Virtual Try-On Systems
- 3D Avatar Generation
- Human Body Reconstruction
- Blender
- Python
- GitHub

Professional Skills

- Technical Research
- Documentation
- Problem Solving
- Comparative Analysis
- Project Organization
- Independent Learning

Key Learnings

One of the most valuable aspects of this internship was understanding how multiple AI technologies work together to create Virtual Try-On systems.

Some of the major learnings include:

- Understanding body landmark detection.
- Learning the importance of body segmentation.
- Exploring garment fitting techniques.
- Understanding human body reconstruction.
- Learning about 3D avatar generation workflows.
- Studying future applications of AR and VR in fashion technology.
- Understanding the role of recommendation systems in personalized shopping.

Challenges Faced

During the internship, several challenges were encountered:

- Understanding complex AI research papers.
- Learning new technologies within limited time.
- Exploring advanced 3D reconstruction concepts.
- Managing and organizing large numbers of digital assets.
- Understanding the workflow of modern Virtual Try-On systems.

These challenges encouraged continuous learning and improved my research and analytical abilities.

Overall Experience

The internship was a valuable learning experience that provided exposure to modern Artificial Intelligence and Computer Vision technologies. It helped me understand how Virtual Try-On systems are developed and how AI can be applied to solve real-world problems in fashion technology and e-commerce.

The combination of research, practical exploration, and documentation activities enhanced both my technical knowledge and professional skills.

Conclusion

The Raritone Summer Internship 2026 provided a strong foundation in AI-powered Virtual Try-On systems, body reconstruction technologies, and digital avatar generation. Through research and exploration, I gained valuable insights into emerging technologies and developed skills that will support my future learning and career growth in Artificial Intelligence and Computer Vision.